



GeoXO Mission Integration Prime: Concept and Alternatives

March 25, 2026

Introduction



- GeoXO is adopting a **Mission Integration Prime** contract to provide launch services, mission operations, and ground systems for GeoXO.
 - Mission Integration Prime (MIP) contract to be awarded in early 2028
- MIP contract scope replaces functions provided by the Government for previous GOES missions:
 - Launch Services - previously NASA Launch Services Program – (LSP)
 - Ground Systems - previously GOES-R Ground Segment / NOAA OSPO
 - Mission Operations - previously Government operations teams
 - L1b Data Operations - previously Government operations teams
- *Preliminary alternatives to the MIP concept have been identified and are presented in this slide package*
 - *Additional alternatives are welcome and may be identified through RFI responses*
- For GeoXO, Spacecraft and Instrument vendor support to MIP should be very similar to support provided previously to Government operations teams and NASA LSP.

Objectives



- Present basis for the GeoXO Mission Integration Prime (MIP) contractor
- Define MIP roles & responsibilities
- Identify MIP alternative concepts

Assumptions for MIP Concept

(to be updated based on results of Analysis of Alternatives)

• S/C Contractor will perform the following:

- All satellite integration and test
- Development of Satellite T&C DB
- Development of spacecraft operations handbook and spacecraft recommended operating procedures
- Development of spacecraft required operations tools (e.g. OPS-09 - CODY, PDB, FMDB)
- Satellite testing and processing at the PPF (MIP will support)
- Handover of satellite to MIP at the PPF (at integration with the LV)
- Spacecraft on-orbit performance validation
- Support LOR and spacecraft PLT
- Supports satellite anomaly response, investigation, and recovery

• Instrument Contractors will perform the following:

- Instrument integration and test support
- Development of Instrument T&C DB
- Development of instrument operations handbook and instrument recommended operating procedures
- Development of instrument required operations tools
- Instrument ground processing algorithm definition
- Instrument on-orbit performance validation
- Support instrument commissioning/PLT and L1b product validation
- Supports satellite anomaly response, investigation, and recovery

• MIP will perform the following:

- Launch vehicle services and payload processing facility procurement
- Satellite to Launch Vehicle integration
- Pre-launch End-to-End, Mission Readiness, and RF Compatibility testing
- Prelaunch operations product development and testing
- MM and PG ground system development
- All satellite command and control from lift-off through end-of-life
- Spacecraft and instrument commissioning/post-launch testing
- Sustaining operations from MIP facility/control center
- L1b product generation and delivery
- L1b product monitoring including INR and radiometric calibration trending
- Leads real-time anomaly response and recovery
- Leads off-line satellite anomaly investigation
- Orbital conjunction analysis and risk assessment

Spacecraft and Instrument Contractor activities are largely unaffected by the addition of a MIP

GeoXO Mission Phases

MIP Roles and Responsibilities

MIP Responsibility (S/C, Inst Support)	Launch MIP Responsibility (S/C, Inst Support)	MIP Responsibility (S/C, Inst Support)	MIP Responsibility	
Pre-Launch	LOR	Post Launch Test (PLT)	Operational	Deactivation
MIP is responsible for command, control, monitoring for entire mission				
Pre-Launch Ground systems development and testing Launch Vehicle and payload processing facility procurement Ops product development and validation T&C DB validation End-to-End testing Operations and launch reviews Operations training Contingency planning Mission and Countdown Rehearsals Post-Launch Test Planning Launch planning and coordination Launch commit criteria	Launch & Orbit Raising (LOR) Launch countdown and ascent operations (LV-provider) S/C contractor controls S/C through liftoff MIP controls S/C following liftoff Separation and initial acquisition Geo station acquisition Collision avoidance and risk assessments (CARA) Mechanism deployments Health and safety monitoring Offline analysis External ground station coordination Instruments off/H&S monitoring	Post Launch Test (PLT) Outgassing Instrument activation Spacecraft PLT Instrument PLT Product PLT (L1b) Mission Planning Autonomous maneuver validation Storage mode validation (TBD) Real-time monitoring Off-line Monitoring CARA and COLA monitoring	Operational Mission Planning Real-time monitoring Offline/remote analysis Autonomous stationkeeping and momentum management Offline trending and reporting CARA and COLA monitoring MESO image commanding L1b product generation, monitoring, and distribution to cloud INR and radiometric trending	Deactivation EOL consumable tracking Decommissioning orbit maneuvers (>300km) Propellant depletion Discharge batteries All subsystems deactivation

MIP Activity Participation Matrix

Activity	Legend ☆ - Leads/Co-Leads ✓ - Supports	Participant			
		MIP	S/C Contr	Instr Contrs	NOAA/ OSPO
Ground System Development and Testing	☆				
Launch Vehicle Procurement	☆				
Payload Processing Facility Procurement	☆				
Launch Base Satellite Processing	✓	☆			
Operations Procedure Development	☆	✓	✓		
End-to-End Testing	☆	✓	✓		
T&C Database Development and CM		☆			
Operations Reviews	☆	✓	✓		
S/C Training		☆			
Instrument Training			☆		
Contingency Planning	☆	✓	✓		
Mission Rehearsals	☆	✓	✓		
Post Launch Test Planning	☆	✓	✓		
External Ground Station Coord	☆				
Mission Readiness Testing (Networks)	☆				
Satellite Launch Commit Criteria		☆			
Master Launch Commit Criteria	☆				
Launch Countdown Direction	☆	✓			

Activity	Participant			
	MIP	S/C Contr	Instr Contrs	NOAA/ OSPO
Launch and Orbit Raising Ops	☆	✓	✓	
On-Orbit Anomaly Response	☆	✓	✓	
Collision Avoidance and Risk Assessment	☆			
On Orbit Health and Safety Monitoring	☆			
S/C Post Launch Testing	☆	✓		
Instrument Post Launch Testing	☆		✓	
Product PLT (L1b)	☆		✓	
On-Orbit anomaly response	☆	✓	✓	
Spacecraft Operations Handbook		☆		
Instrument Operations Handbook			☆	
On-Station Operations	☆			
L1b Product Generation and Distribution	☆			
GPA Development			☆	
GPA Implementation and Validation	☆		✓	
Mission Planning	☆			
MESO Image Commanding	☆			
INR and Radiometric Trending	☆			
DCS Operations (DCPR & DCDC)				☆

MIP Contract Structure



- MIP contract will include Launch Vehicle, Payload Processing Facility, and all Mission Operations through 10 years of on-orbit operation
- NASA acquisition
- Base contract includes GeoXO-1 and GeoXO-2 only
 - GeoXO-3 and GeoXO-4 may be options
- Firm Fixed-Price with performance incentive CLINs

MIP Alternatives

- MIP concept is still in formulation
- Preliminary alternatives to the MIP Concept have been identified
- RFI will be released in March 2026 to solicit industry capabilities and feedback to the MIP approach
 - Additional alternatives may be identified through industry feedback
- Analysis of Alternatives to be performed to determine the optimal MIP approach

1. MIP Alternatives – Ground Systems and Operations

A. *MIP Contractor provides all ground system resources*

- MIP provides all ground systems (Mission Management, Antennas, Product Generation, Operations Facility)
- MIP operates the GeoXO satellites from launch through end of mission from MIP facility
- MIP provides L0 and L1b data to NESDIS Cloud

B. *MIP provides Mission Management and Product Generation ground system resources; Uplink/Downlink at WCDAS/CBU*

- MIP provides ground systems for Mission Management, Product Generation, and Operations Facility
- NOAA/OSPO provides antennas at WCDAS, CBU
- Telemetry and commands routed through NOAA/OSPO antennas
- MIP operates the GeoXO satellites from launch through end of mission from MIP facility
- MIP provides L0 and L1b product data to the NESDIS Cloud

C. *MIP provides Mission Management ground system; Uplink/Downlink and L1b product generation by NOAA/OSPO*

- MIP provides ground system resources for Mission Management and Operations Facility
- NOAA provides antennas at WCDAS and CBU
- Telemetry and commands routed through NOAA/OSPO antennas
- NOAA performs L1b Product Generation and delivers L0 and L1b to the NESDIS Cloud
- MIP operates the GeoXO satellites from launch through end of mission from MIP facility

D. *MIP utilizes NOAA GEO ground system*

- NOAA/OSPO provides all ground systems (Mission Management, Antennas, Product Generation, NOAA Facilities)
- MIP operates the GeoXO satellites from launch through end of mission from NOAA Facility & Ground System
- MIP provides L0 and L1b data to the NESDIS Cloud

E. *MIP utilizes NOAA GEO ground system through PLT; Sustaining Operations by NOAA/OSPO*

- NOAA/OSPO provides all ground systems (Mission Management, Antennas, Product Generation, NOAA Facilities)
- MIP operates the GeoXO satellites from launch through end of PLT
- Handover to NOAA/OSPO occurs following spacecraft and instrument activation and completion of post-launch test
- NOAA/OSPO performs sustaining operations for the mission life

MIP Alternatives – Launch Services



2. MIP Alternatives – Launch Services

- A. *MIP procures all launch services (Payload Processing Facility and Launch Vehicle) commercially*
- B. *NASA/NOAA procures launch services using LSP contract vehicles*
- C. *NASA/NOAA directly procures launch services commercially*

MIP Schedule



- MIP RFI Release March 2026
- Government Analysis of Alternatives complete Summer 2026
 - Informed by RFI responses
- Draft RFP Release, Early 2027
- Final RFP Release, Summer 2027
- MIP Award, Early 2028